DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 6 (AT3)

Case Study

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Case Study 1 – Retail Sales Performance Dashboard

1. Scenario

A retail company has collected one year of sales data from five regions: North, South, East, West, and Central.
The dataset contains monthly sales, product categories, customer segments, advertising expenditure, and total revenue.
Management wants to identify:

- The highest-performing region.
- Monthly sales trends.
- Revenue contribution of each product category.
- Relationship between advertising expenditure and sales.
- Sales distribution across regions.

2. Selection of Suitable Visualizations

Requirement	Suitable Visualization	Purpose
Regional revenue comparison	Bar Plot	Compare revenue across five regions.
Monthly sales trend	Stacked Bar Chart	Compare monthly sales by region.
Product category contribution	Stacked Bar Chart	Show category-wise revenue contribution.
Advertising expenditure vs sales	Scatter Plot	Identify the relationship between two variables.
Regional sales distribution	Histogram and Density Plot	Understand the spread and distribution of sales.

3. Dataset

The following R program generates a simulated one-year retail sales dataset containing 60 observations (12 months × 5 regions).

4. R Programming – Complete Implementation

Install and load the required package

```
install.packages("ggplot2")  # Run once if not installed
library(ggplot2)
```

Dataset creation

```
# Generate one year of retail sales data
set.seed(123)

retail <- expand.grid(
  Month = 1:12,
  Region = c("North", "South", "East", "West", "Central")
)
```

```
# Add product categories and customer segments
retail$Category <- rep(
  c("Electronics", "Clothing", "Grocery", "Furniture"),
  length.out = 60
)

retail$Segment <- rep(
  c("Regular", "Premium", "New"),
  length.out = 60
)

# Generate advertising expenditure
region_ad <- c(
  North = 1000, South = 500, East = 1500,
  West = 800, Central = 1200
)

retail$Advertising <- 8000 +
  retail$Month * 500 +
  region_ad[retail$Region]

# Generate sales revenue
region_base <- c(
  North = 100000, South = 90000, East = 110000,
  West = 95000, Central = 105000
)

category_bonus <- c(
  Electronics = 20000, Clothing = 10000,
  Grocery = 5000, Furniture = 15000
)

retail$Revenue <- round(
  region_base[retail$Region] +
  retail$Month * 2500 +
  retail$Advertising * 3 +
  category_bonus[retail$Category] +
  sin(1:60) * 5000
)

# Display the dataset
print(retail)

# Save the dataset
write.csv(retail, "Retail_Sales_Dataset.csv", row.names = FALSE)
```

Q1. Bar Plot – Regional Revenue Comparison

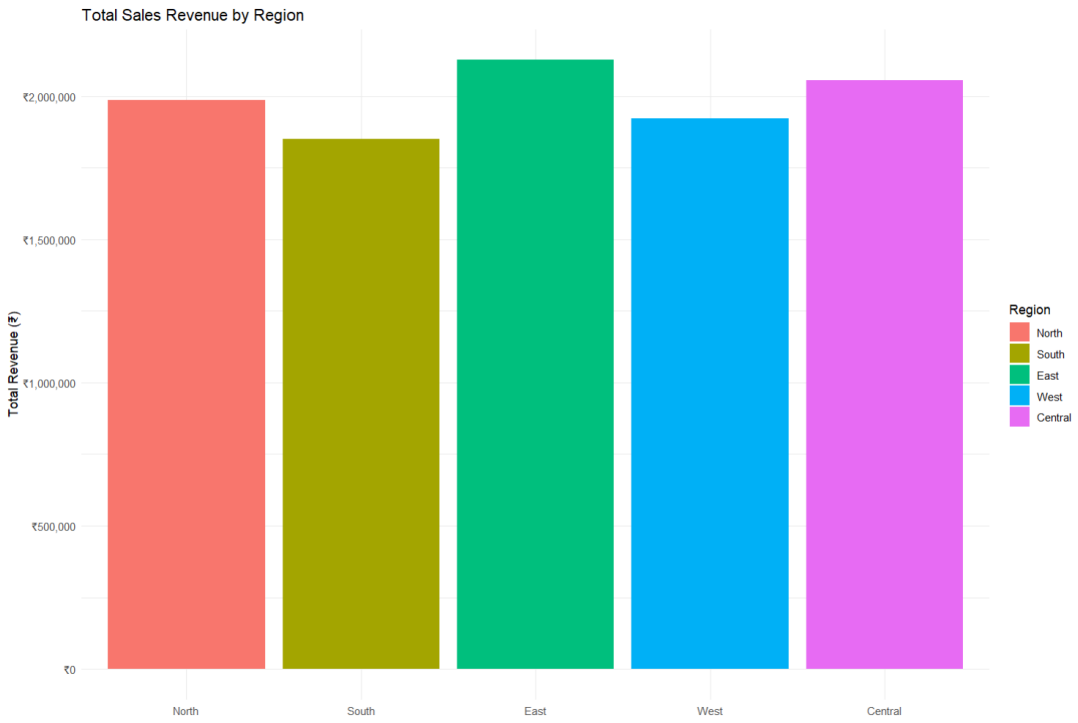
Aim: Compare the total sales revenue generated by the five regions.

```
# Calculate total revenue by region
region_sales <- aggregate(
  Revenue ~ Region,
  data = retail,
  sum
)

# Create bar plot
p1 <- ggplot(region_sales,
             aes(x = Region, y = Revenue, fill = Region)) +
  geom_col() +
  labs(
    title = "Total Sales Revenue by Region",
    x = "Region",
    y = "Total Revenue (₹)",
    fill = "Region"
  ) +
  scale_y_continuous(labels = scales::label_comma(prefix = "₹")) +
  theme_minimal()

print(p1)

# Save the chart
ggsave("Bar_Plot.png", p1, width = 8, height = 6, dpi = 300)
```



Interpretation:

- The bar plot compares the total revenue of all five regions.
- The tallest bar represents the highest-performing region.
- The shortest bar represents the lowest-performing region.

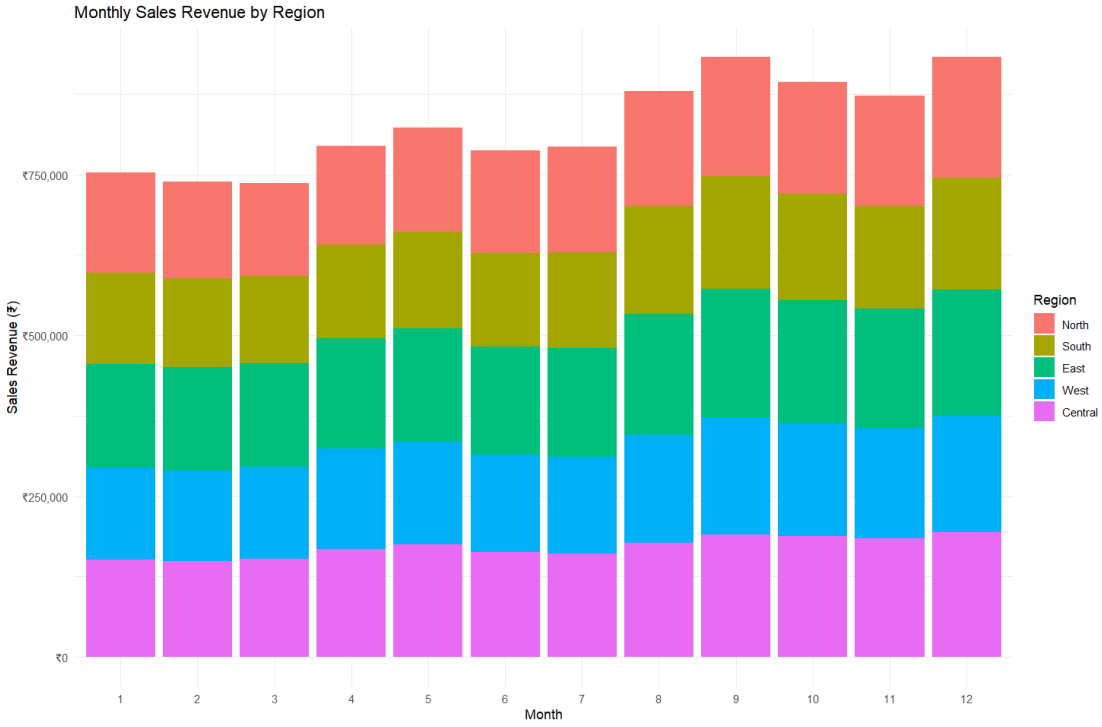
Q2. Stacked Bar Chart – Monthly Sales by Region

Aim: Compare monthly sales revenue contributed by each region over one year.

```
# Create stacked bar chart
p2 <- ggplot(retail,
             aes(x = factor(Month), y = Revenue, fill = Region)) +
  geom_col() +
  labs(
    title = "Monthly Sales Revenue by Region",
    x = "Month",
    y = "Sales Revenue (₹)",
    fill = "Region"
  ) +
  scale_y_continuous(labels = scales::label_comma(prefix = "₹")) +
  theme_minimal()

print(p2)

# Save the chart
ggsave("Stacked_Bar_Chart.png", p2,
       width = 9, height = 6, dpi = 300)
```



Interpretation:

- Each bar represents the total revenue generated in one month.
- Different colours represent the five regions.
- The chart helps compare monthly sales and each region's contribution.

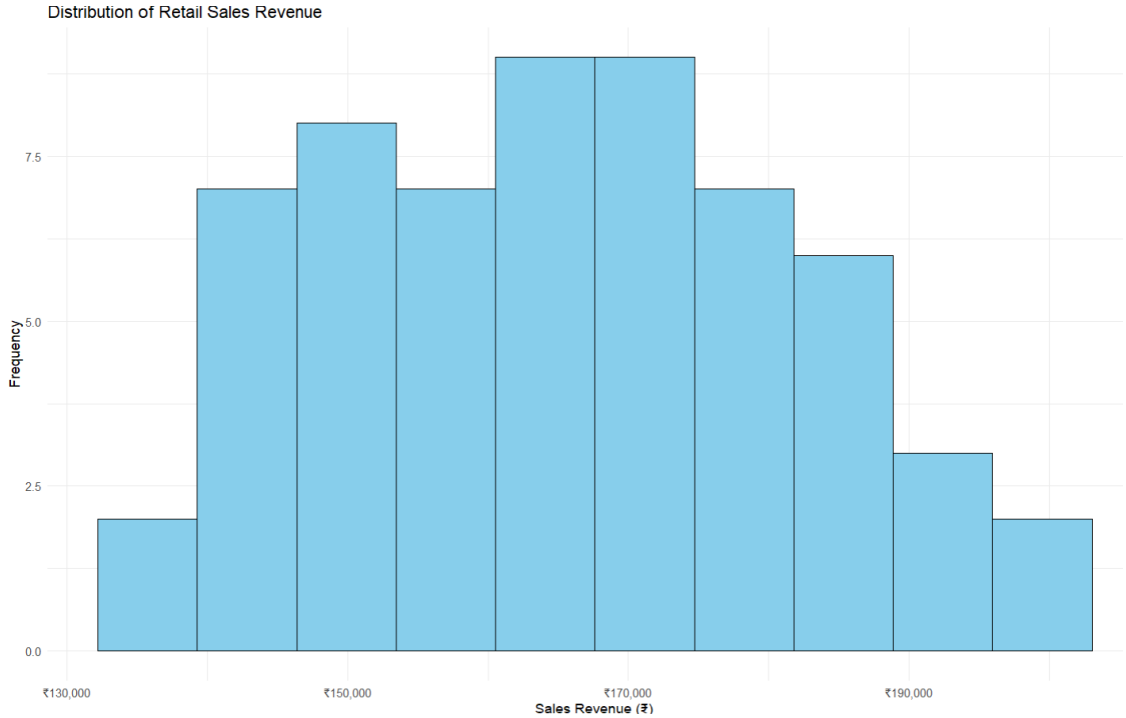
Q3. Histogram – Sales Revenue Distribution

Aim: Display the frequency distribution of sales revenue.

```
# Create histogram
p3 <- ggplot(retail, aes(x = Revenue)) +
  geom_histogram(
    bins = 10,
    fill = "skyblue",
    color = "black"
  ) +
  labs(
    title = "Distribution of Retail Sales Revenue",
    x = "Sales Revenue (₹)",
    y = "Frequency"
  ) +
  scale_x_continuous(labels = scales::label_comma(prefix = "₹")) +
  theme_minimal()

print(p3)
```

```
# Save the chart
ggsave("Histogram.png", p3, width = 8, height = 6, dpi = 300)
```



Interpretation:

- The histogram groups sales revenue into different intervals.
- Taller bars indicate more observations within an interval.
- It helps understand the spread and frequency of sales revenue.

Q4. Density Plot – Regional Sales Distribution

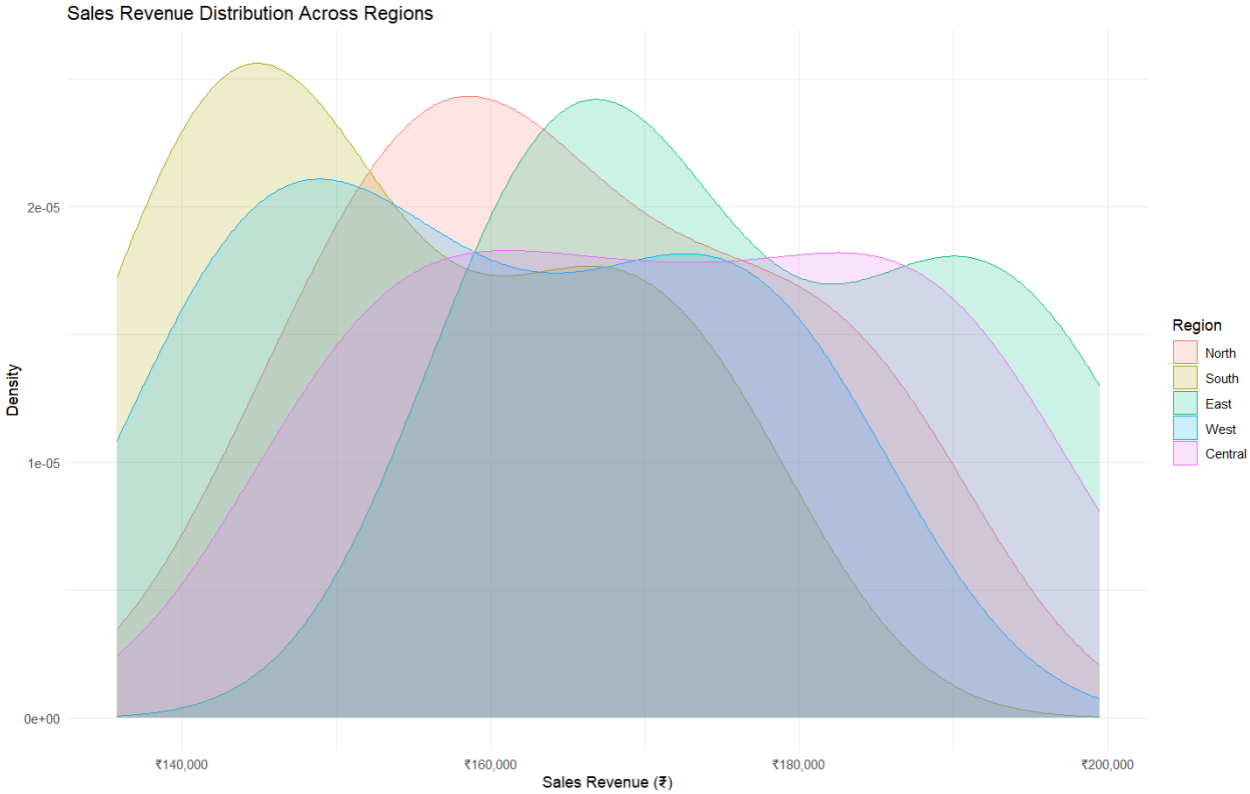
Aim: Compare the distribution of sales revenue across the five regions using density curves.

```
# Create density plot
p4 <- ggplot(retail,
             aes(x = Revenue, color = Region, fill = Region)) +
  geom_density(alpha = 0.2) +
  labs(
    title = "Sales Revenue Distribution Across Regions",
    x = "Sales Revenue (₹)",
    y = "Density",
    color = "Region",
    fill = "Region"
  ) +
  scale_x_continuous(labels = scales::label_comma(prefix = "₹")) +
  theme_minimal()

print(p4)

# Save the chart
```

ggsave("Density_Plot.png", p4, width = 8, height = 6, dpi = 300)



Interpretation:

- Each coloured curve represents one region.
- The peaks show where sales values are concentrated.
- The curves help compare the distribution and variation of sales across regions.

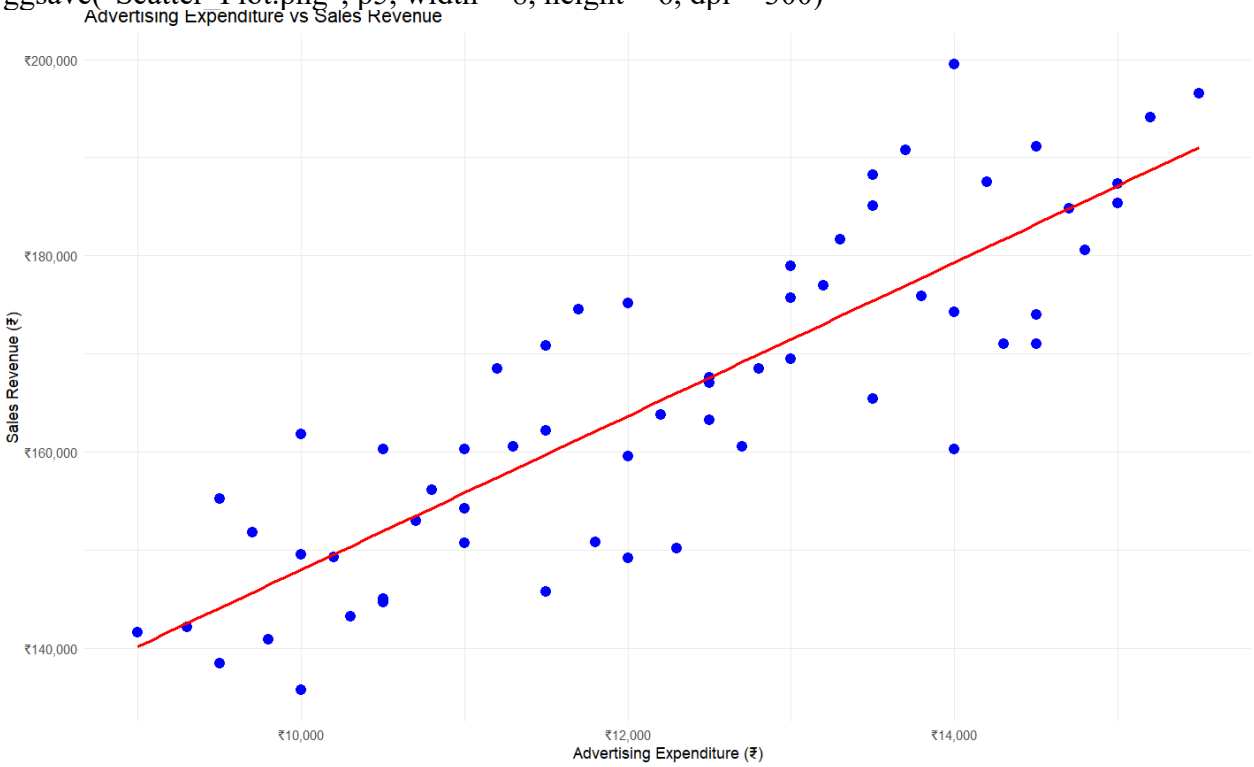
Q5. Scatter Plot – Advertising Expenditure vs Sales

Aim: Identify the relationship between advertising expenditure and sales revenue.

```
# Create scatter plot with fitted trend line
p5 <- ggplot(retail,
             aes(x = Advertising, y = Revenue)) +
  geom_point(color = "blue", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "Advertising Expenditure vs Sales Revenue",
    x = "Advertising Expenditure (₹)",
    y = "Sales Revenue (₹)"
  ) +
  scale_x_continuous(labels = scales::label_comma(prefix = "₹")) +
  scale_y_continuous(labels = scales::label_comma(prefix = "₹")) +
  theme_minimal()

print(p5)
```

Save the chart
ggsave("Scatter_Plot.png", p5, width = 8, height = 6, dpi = 300)



Interpretation:

- Each point represents one region's sales observation for a month.
- The red line represents the fitted linear trend.
- An upward-sloping line indicates a positive relationship between advertising expenditure and sales revenue.

5. Final Analysis and Interpretation

The following results are obtained from the simulated dataset generated by the program.

1. Which region generated the highest revenue?

Region	Total Revenue
North	₹19,85,373
South	₹18,48,997
East	₹21,26,312
West	₹19,21,704
Central	₹20,55,787

Answer: East region generated the highest revenue of ₹21,26,312, while South generated the lowest revenue.

2. Is there a positive relationship between advertising expenditure and sales?

Answer: Yes. The scatter plot shows an upward trend. As advertising expenditure increases, sales revenue generally increases. This indicates a positive relationship in the simulated dataset.

3. Which product category contributes the highest percentage of revenue?

Product Category	Revenue
Electronics	₹25,08,606
Clothing	₹24,20,367
Grocery	₹24,02,195
Furniture	₹26,07,005

Answer: Furniture contributes the highest revenue, approximately 26.6% of total revenue.

6. Additional Stacked Bar Chart – Product Category Contribution

To show product category contribution clearly, use this stacked bar chart. The monthly sales trend can be displayed separately using the line chart below.

```
# Stacked bar chart showing category revenue by region

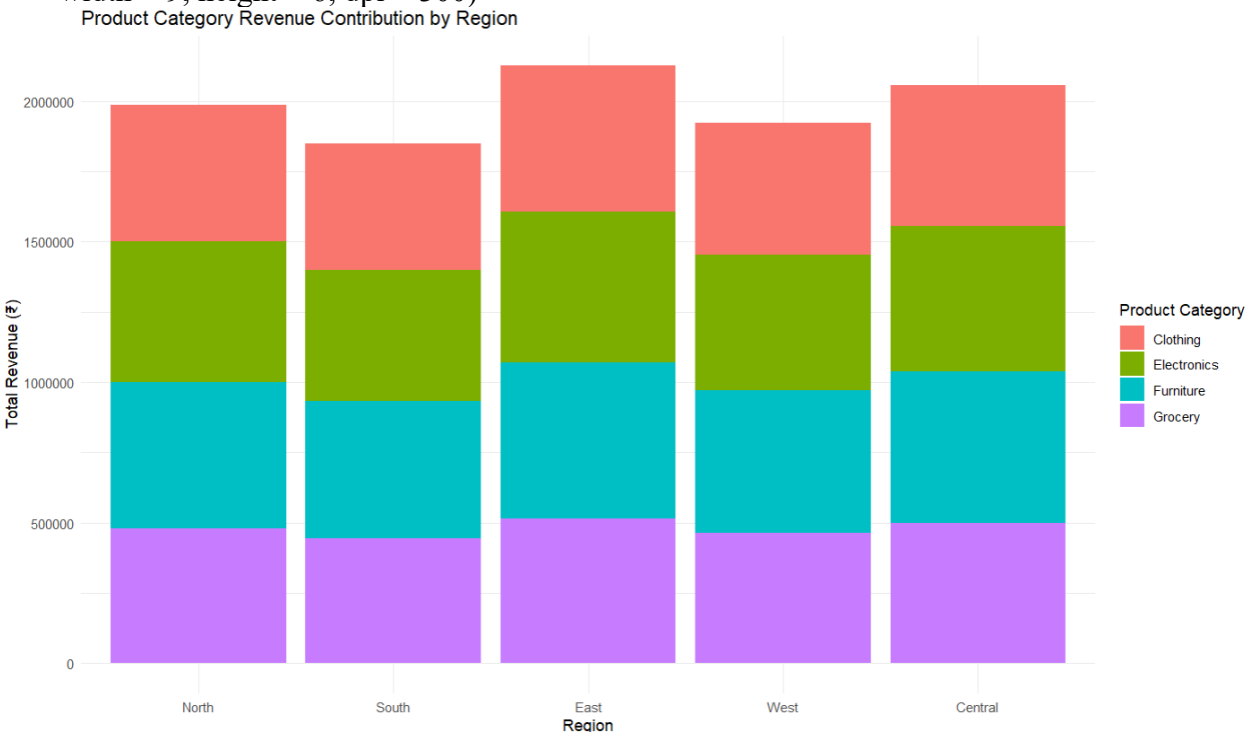
category_sales <- aggregate(
  Revenue ~ Region + Category,
  data = retail,
  sum
)

p6 <- ggplot(
  category_sales,
  aes(x = Region, y = Revenue, fill = Category)
) +
  geom_col() +
  labs(
    title = "Product Category Revenue Contribution by Region",
    x = "Region",
    y = "Total Revenue (₹)",
    fill = "Product Category"
  ) +
  theme_minimal()

print(p6)

ggsave("Category_Contribution.png", p6,
```

width = 9, height = 6, dpi = 300)



7. Additional Line Chart – Monthly Sales Trend

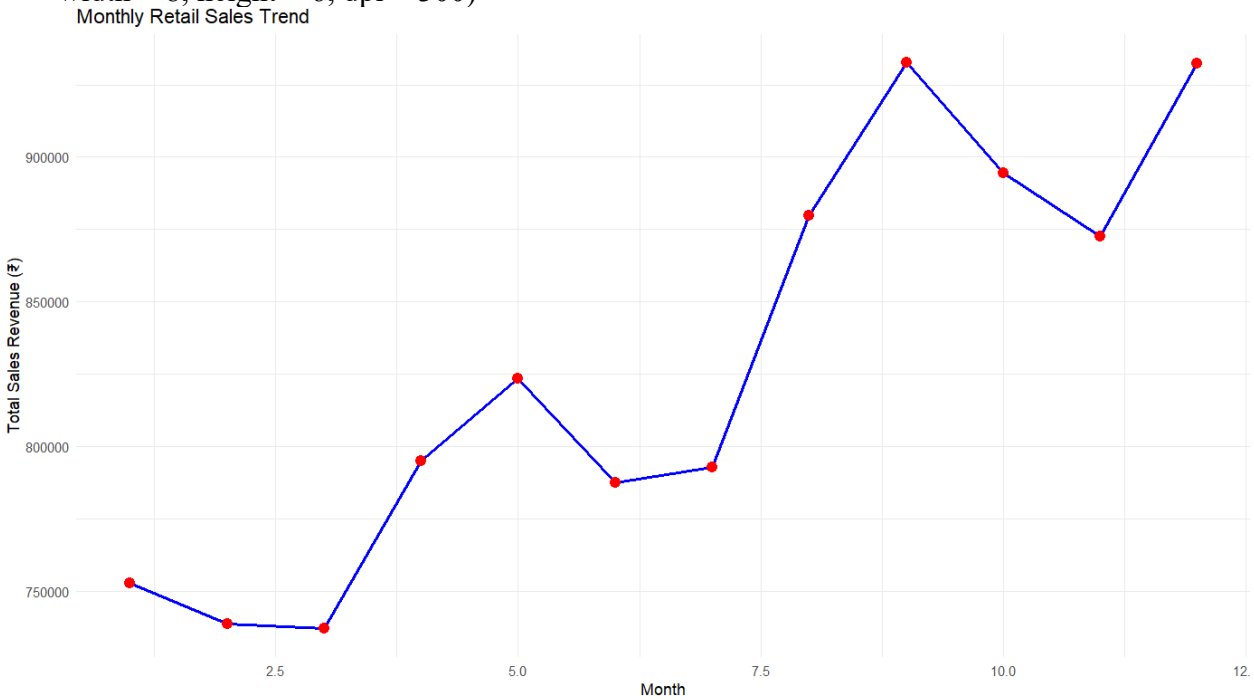
```
# Calculate monthly total sales
monthly_sales <- aggregate(
  Revenue ~ Month,
  data = retail,
  sum
)

# Create line chart
p7 <- ggplot(monthly_sales, aes(x = Month, y = Revenue)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Monthly Retail Sales Trend",
    x = "Month",
    y = "Total Sales Revenue (₹)"
  ) +
  theme_minimal()

print(p7)

ggsave("Monthly_Sales_Trend.png", p7,
```

width = 8, height = 6, dpi = 300)



Interpretation: The line chart shows the monthly changes in total sales revenue. It helps management identify increasing or decreasing sales trends throughout the year.

8. Overall Conclusion

The retail sales dashboard helps management understand business performance through different visualizations.

- East region generated the highest revenue.
- Furniture contributed the highest percentage of total revenue.
- Advertising expenditure and sales revenue show a positive relationship.
- Histograms and density plots help understand sales distribution.
- Stacked bar charts and line charts help compare category contributions and monthly sales trends.

Thus, data visualization helps the retail company identify business performance, understand sales patterns, and make better decisions.